ADVANCED BUSINESS ANALYTICS · FINAL REPORT · OCTOBER 2026

Vietnam Real Estate AVM

An Automated Valuation and Market Intelligence Platform for Hanoi and Ho Chi Minh City

Team: Advanced Business Analytics capstone group **Date:** 15 October 2026

Evidence: 5,268 homes listed for sale on mogi.vn, matched to 34,321 OpenStreetMap points. Every accuracy figure is measured on homes the models had not seen.

60% to 64%
apartments and alley houses valued within ±20%

1.5 times
stale rate in the top 10% of risk scores vs average

VND 9.7 bn
net partner value a year, base case

EXECUTIVE SUMMARY

The answer. A valuation platform built from public listings and open map data gives brokers in Hanoi and Ho Chi Minh City an independent fair value for secondary-market homes. On homes it had not seen, it valued 60% of apartments and 64% of alley houses within ±20% of the asking price. It is ready for a broker pilot on these two types, not yet for lending.

Situation. Homes in Hanoi and HCMC trade for billions of VND each. Yet buyers, brokers and lenders price secondary-market homes with no independent reference, so the asking price anchors every negotiation.

Complication. Existing tools cover listed apartment projects (OneHousing) or sit inside banks (Citics). Alley houses, street-front houses, villas and land make up most of the secondary market, and none has a public valuation.

Resolution. The platform joins 5,268 mogi.vn sale listings to 34,321 OpenStreetMap points and gives three answers. The **Fair Value Engine** says what a home is worth, with a 90% range and comparables. The **Stale Listing Radar** flags listings likely to stay unsold for more than 180 days. The **Market Map** shows how districts differ in price, speed and housing mix.

Five findings

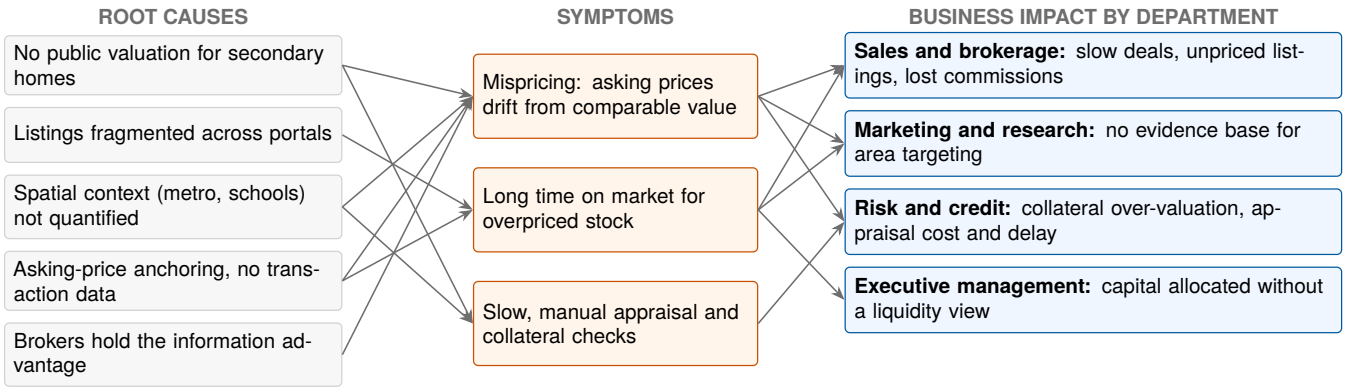
1. **The engine is good enough for broker use on apartments and alley houses:** 60% and 64% fall within ±20% (53.9% across all types). Half of all estimates miss by more than 18.1%, so every estimate comes with a range.
2. **Street-front houses, land and lending need more work.** Only 42% of street-front houses and 43% of land plots fall within ±20%, and expensive homes tend to be under-valued.
3. **Map data stands in for local knowledge.** Distances to the centre, metro, schools and markets deliver 60% of the accuracy gain from knowing the district, so thinly listed districts can still be valued.
4. **Stale risk can be ranked.** The radar reaches an AUC of 0.81 (0.5 is a coin flip, 1.0 a perfect ranking). Its top 10% of listings are stale 1.5 times as often as average (99% against 65%).
5. **Stale listings are not mainly overpriced listings.** The price gap ranks 12 of 26 drivers, and a rule built on it is no better than a coin flip; staleness tracks the district and the type of seller. Brokers should use the radar to target effort and the price gap to coach sellers.

Expected value. In the base case the platform creates net partner value of VND 9.7 billion a year, 4.4 times its run cost of VND 2.2 billion. Scenarios range from a loss of VND 2.3 billion to a gain of VND 40.9 billion. Broker adoption moves the result more than any accuracy metric.

Recommendation. Pilot the Fair Value Engine with one brokerage on apartments and alley houses, always with its range. Use the radar as a weekly review list. Keep other types on desk review, and calibrate on transaction prices before any lending use.

Roadmap. *Pilot (months 1 to 3):* weekly data collection, a one-agency broker pilot, a monthly accuracy report. *Scale (months 4 to 7):* a second portal, transaction prices, a bank pre-screening trial. *Productise (months 8 to 12):* a public valuation page and API, a quarterly market index. Each phase ends with a go or no-go gate: accuracy confirmed on pilot deals, bank sign-off, public launch.

Exhibit 1. Five root causes lead to mispricing, long time on market and slow appraisal, with a different cost in each department



Source: team analysis; desk research in Section 2.

Exhibit 2. Existing tools serve project apartments or banks; no public tool values townhouses, villas and land in the secondary market

Solution	Serves	Coverage (as reported)	Method	Access	Gap this platform addresses
OneHousing (One Mount)	Buyers of project homes; Techcombank borrowers	533,856 homes in 627 projects one month after the 2022 launch, mostly high-rise apartments (Báo Pháp luật 2022); company-reported error below 5% (VnEconomy 2022)	Proprietary model on developer and transaction data	Free lookup; paid valuation note accepted by Techcombank (Techcombank 2023)	Listed projects only; no alley houses, no land, no published method
Citics Valuation	Banks	Used by more than 32 banks; over 33 million properties digitised in 38 provinces; USD 2.1 million raised in 2025 (e27 2025)	Automated valuation inside the bank workflow	Institutions only	No buyer or broker access
Rever	Buyers and sellers via brokers	Brokerage with a technology layer (Brands Vietnam 2021)	Broker comparables	Through agents	No public valuation
Propzy	Brokerage platform	Closed in September 2022 after failing to raise capital (VietnamFinance 2022)	Broker comparables	Closed	Shows the cost of a capital-heavy model
Bank appraisal	Lenders	Every mortgage file	Site visit and comparables	Internal	Slow and costly per file
Broker judgement	Most secondary deals	All types	Personal comparables	Informal	No error band, no audit trail

Source: company pages and press reports cited in the table (figures as reported by the companies, not independently verified); team analysis.

THE PROBLEM: AN OPAQUE SECONDARY MARKET

STRATEGIC PROBLEM STATEMENT

Buyers, independent brokers and lenders in Hanoi and Ho Chi Minh City set and accept prices for secondary-market homes without an independent reference. Existing valuation products cover listed apartment projects or serve banks only. Most townhouse, villa and land deals are therefore priced by anchoring on the asking price. The platform answers three questions: what is this home worth, is the asking price out of line, and which districts move slowly?

The same missing number, a fair value with a stated error, serves four groups that make different decisions (Exhibit 3). This makes the problem cross-functional: one tool, four uses.

Two links carry most of the cost (Exhibit 1). First, without a public valuation, asking prices anchor negotiations and drift from the value of comparable homes. Brokers then lose time on listings that do not sell, and banks risk over-valuing collateral. Second, experienced brokers see the value of a metro station, a school or a short commute, but buyers cannot measure it. That gap preserves the broker's information advantage and slows every deal. Section 6 returns to the same four departments with evidence from the platform.

MARKET SCAN AND EVIDENCE BASE

The public, all-type quadrant is empty. OneHousing is public but covers listed apartment projects. Citics covers all types but serves banks only (Exhibits 2 and 5). Rever and Propzy built brokerages, not valuation tools, and Propzy closed in 2022. A buyer or broker pricing an alley house, a villa or a plot of land has no independent number to check.

Exhibit 3. Each stakeholder needs the same fair-value reference for a different decision

Stakeholder	Decision	What is missing today
Buyers	Offer price, walk away or negotiate	No reference outside the listing itself
Brokers (sales)	List price advice, which sellers to coach	Comparables found by hand, no error band
Banks (risk, credit)	Loan amount at 70% loan-to-value	Manual appraisal, slow and costly
Research, marketing	Which districts to target, with what message	No district view of price and speed
Executives	Where to open offices and commit capital	No combined price and speed map

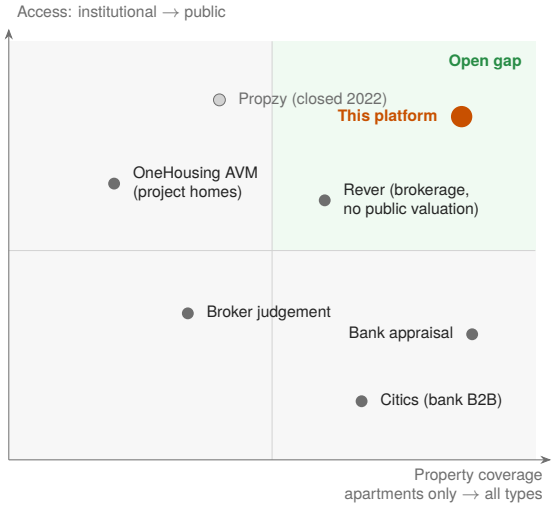
Source: team analysis.

Exhibit 4. Research and practice shape every design choice: a transparent baseline, measured location, a stronger model and accuracy reported the way lenders read it

Question	What research and practice say	Design choice
What drives housing prices?	A home is a bundle of attributes with implicit prices (Rosen 1974). Floor area, rooms, age and location recur across studies with consistent signs (Sirmans et al. 2005). A log-price model with location effects is the standard starting point (Malpezzi 2003).	A transparent regression on log asking price with district effects is the baseline of the Fair Value Engine.
Does transport access raise prices?	Shorter distance to a new London rail station was associated with higher prices (Gibbons and Machin 2005); station effects vary widely across cities (Debrezion et al. 2007). Hanoi runs two urban rail lines; HCMC Line 1 opened on 22 December 2024 (Việt Nam News 2024).	Distance to the nearest operating station enters the engine. Its association with price is measured, not assumed.
Do machine-learning models beat regression?	On 123,000 Swiss sales, gradient boosting gave the most accurate valuations of six methods (Mayer et al. 2019). XGBoost is a fast, regularised version (Chen and Guestrin 2016); SHAP values explain each estimate (Lundberg and Lee 2017).	Gradient boosting competes with the regression on the same unseen homes; SHAP charts explain it to managers.
How do practitioners judge accuracy?	The IAAO standards ask for testing on held-out sales, hit rates within a tolerance and ratio statistics (International Association of Assessing Officers 2018, 2013). Prediction intervals with a coverage guarantee are available without distribution assumptions (Lei et al. 2018).	Accuracy is reported as the share of estimates within ±10% and ±20% (PPE10, PPE20), with a 90% range, by property type.

Source: studies, standards and company sources cited in the table; team analysis.

Exhibit 5. No public tool values every property type



Source: desk research (Exhibit 2); positions are team judgements.

Where this platform differs. It values all property types, including alley houses and land. It publishes its method and its error for each type. It adds measured location data from OpenStreetMap to the listings. It also ranks listings by the risk of staying unsold and maps the districts for managers.

Research and practice settle four design questions (Exhibit 4). Academic studies explain what drives prices and which methods value homes best. Practice adds two lessons. Zillow publishes its error by market segment, and the off-market error is several times the on-market error (Zillow 2026). Zillow also closed its home-buying arm in 2021 after its management said that forecasting home prices was harder than expected (Inman 2021). A valuation is a reference for a human decision, not an automatic bid.

DATA: TWO SOURCES, ONE PROPERTY TABLE

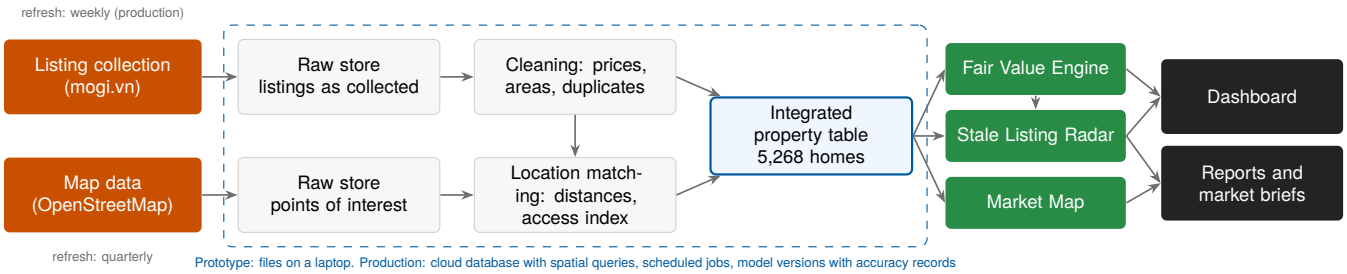
One property table feeds all three engines (Exhibit 6). Listings from mogi.vn supply prices and home attributes. OpenStreetMap supplies the location context: metro stations, schools, hospitals, parks, markets and water. Both are cleaned, matched by map position and stored as one table.

The sample is a prototype sized for a 1.5-hour collection window. The collector gathered 6,000 sale listings with their detail pages in 46 minutes on 15 October 2026: 7,792 listings an hour, with no blocked requests. Quotas per city were set by type, so type shares reflect the quotas, not the market (Exhibit 8). Hanoi apartments ran out after about 675 listings, and the shortfall moved to other types. The site lists newest first, so a fifth of each quota came from the last result pages to include older listings. A scheduled run covers the full market of at least 33,000 listings in about 5 hours.

Each home carries four groups of facts (Exhibit 7). Physical facts: area, rooms, storeys, frontage and access road width. Legal status. Text flags from the description, such as car access, corner lot or business use. Location: distances to the centre and to the nearest metro station, school, hospital, park and market, plus one accessibility score. Missing values are filled with the typical value for the type and flagged, so the engines can learn whether a gap itself carries information.

Cleaning kept 88% of the listings (Exhibit 9). The largest loss is the same home posted by several brokers (219). Areas out of range (132), buildings with more than 15 bedrooms sold as serviced apartments (122) and placeholder or mistyped prices (114) follow. The price reader matches the site's own numeric price on 100.0% of listings.

Exhibit 6. Listings and map data are cleaned, matched by location and stored as one property table that serves all three engines



Source: team design.

Exhibit 7. Example extract: eight clean listings across both cities and four property types

ID	City	District (as listed)	Type (site label)	Price (VND bn)	Area (m^2)	VND m/m^2	Beds	Legal	Posted
21710182	TP. Hồ Chí Minh	Quận 10	Căn hộ chung cư	3.50	50	70.0	1	Khác	14/10/2026
22751445	TP. Hồ Chí Minh	Quận Gò Vấp	Nhà hẻm/ngõ	4.25	40	106.2	3	Sổ hồng	09/10/2026
22715805	TP. Hồ Chí Minh	Quận 1	Nhà mặt tiền/phố	35.00	71	493.0	13	Sổ hồng	15/10/2026
22726003	TP. Hồ Chí Minh	Quận 12	Đất	5.19	92	56.4	0	Sổ hồng	12/10/2026
22673730	Hà Nội	Quận Cầu Giấy	Căn hộ chung cư	9.72	136	71.5	3	Sổ đỏ	18/04/2025
22747097	Hà Nội	Quận Đống Đa	Nhà hẻm/ngõ	12.50	50	250.0	6	Sổ đỏ	16/08/2026
22415247	Hà Nội	Quận Đống Đa	Nhà mặt tiền/phố	49.90	131	380.9	8	Sổ đỏ	14/08/2026
22723649	Hà Nội	Quận Tây Hồ	Đất	31.80	151	210.6	0	Sổ đỏ	31/12/2025

Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); prices are asking prices; VND m/m^2 = million VND per square metre.

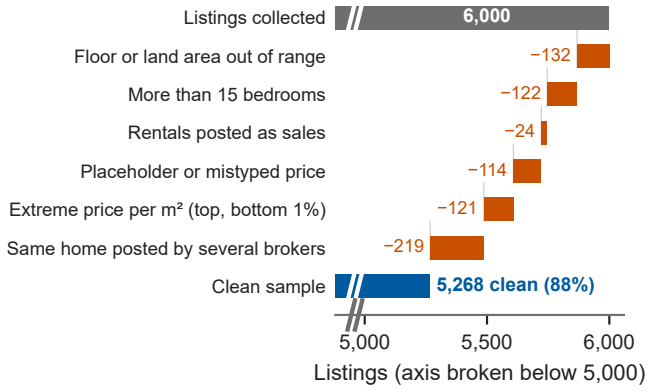
Exhibit 8. Listings supply prices and attributes; map data supplies location; every type has enough homes to report accuracy

	mogi.vn listings	OpenStreetMap
Content	Asking price, area, rooms, legal status, address, map point, text	34,321 points: 20 + 14 metro stations, 11,643 bus stops, schools, hospitals, parks, markets, water
Refresh	Weekly in production	Quarterly
Licence	Public pages, research use	Open Database Licence
Sample	6,000 collected, 5,268 clean: 2,638 Hanoi, 2,630 HCMC; 51 districts	9 layers, 2 cities

Clean homes by type: apartments 1,375, alley houses 1,500, street-front houses 929, villas and terraced 589, land 875

Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); OpenStreetMap contributors (ODbL).

Exhibit 9. 88% of collected listings survive cleaning; duplicate postings remove the most



Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); team analysis.

Integration challenges

- **Location precision.** 27% of listings carry the poster's own map pin; 71% share a pin with listings on other streets, so their distances are accurate to the ward, not the plot.
- **Administrative reform.** Since 1 July 2025 Vietnam has no districts and many wards were merged (PwC Vietnam 2025). The site still uses the old names, so the platform does too.
- **Asking, not transaction, prices.** The engine estimates what comparable sellers ask. Calibration on transaction prices is part of the roadmap (Section 9).
- **Refreshed dates.** Sellers re-post listings, which resets the posted date. Listing age is therefore measured from the creation date implied by the listing ID.

Privacy and ethics

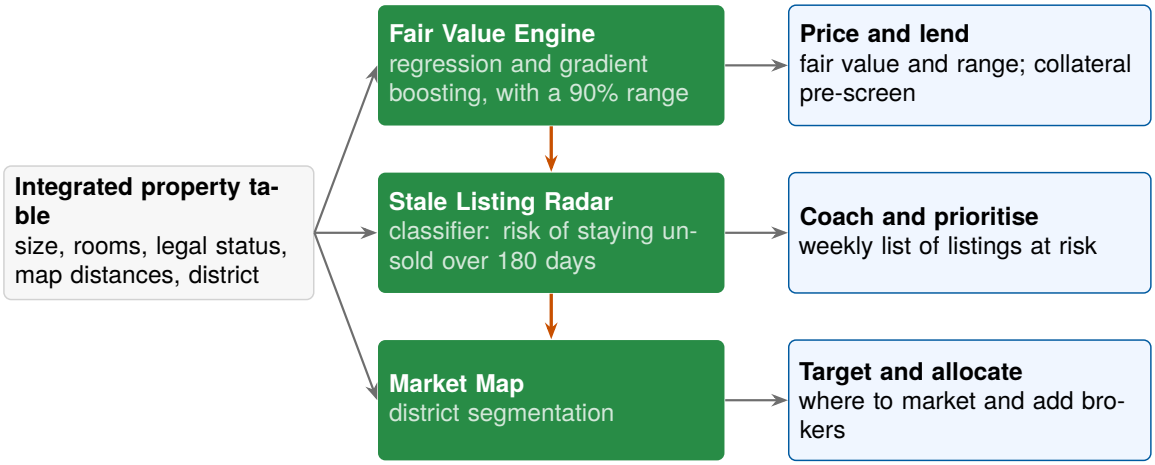
- Only public listing pages were collected, on paths that the site's robots.txt allows; requests were paced and identified as research.
- Phone numbers and e-mail addresses were removed before saving; no agent names or profile links were stored.
- OpenStreetMap data is used under the Open Database Licence with attribution.

HOW THE PLATFORM WORKS

Three engines share one table, and each ends in a decision owned by a department (Exhibit 10). The **Fair Value Engine** combines a transparent hedonic regression with a gradient-boosting valuation model and shows a 90% range around each estimate. The **Stale Listing Radar** is a classifier that ranks listings by the risk of staying unsold for more than 180 days. It uses the gap between the asking price and the fair value as one of its inputs. The **Market Map** groups districts by price, speed, access and housing mix.

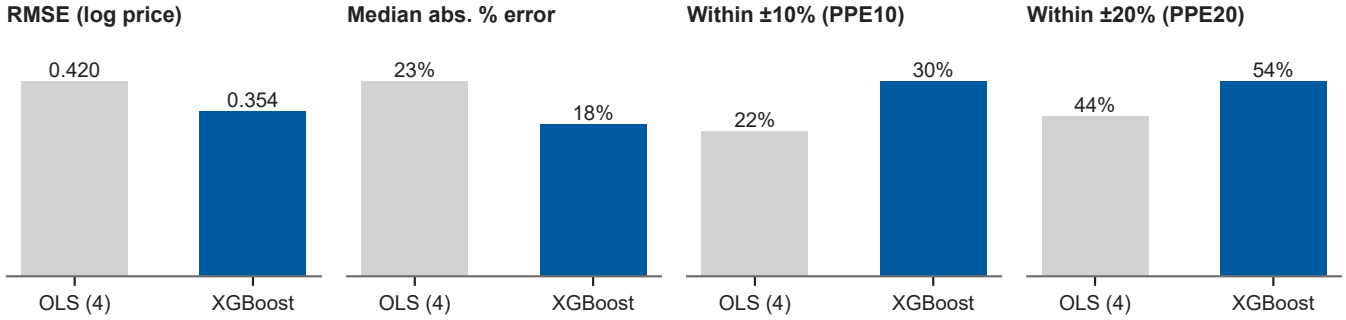
Each engine has an accuracy target for broker use. The regression should explain more than 75% of the vari-

Exhibit 10. Three linked engines turn one property table into three decisions: price and lend, coach and prioritise, target and allocate



Source: team design.

Exhibit 11. The gradient-boosting model beats the regression on every accuracy measure: 53.9% of estimates fall within ±20% of the asking price



Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); OpenStreetMap contributors (ODbL); 1,054 test homes not used to build the models, stratified by city and type; team analysis.

ation in asking prices (R^2 above 0.75). The valuation model should reach a typical error below 0.20 on the log scale (RMSE). For lenders, the same error is restated as the share of estimates within ±10% and ±20% of the asking price (PPE10, PPE20). The radar should rank stale listings with an AUC above 0.72, where 0.5 is a coin flip and 1.0 a perfect ranking. The district segments should be distinct enough to target separately.

How the results were kept honest

Every accuracy figure comes from homes the engines had not seen while they were built. The radar never sees listing age, dates or anything that would reveal the answer. Listing age is measured from the creation date, because sellers refresh the posted date when they re-post (Appendix A1).

RESULTS

Fair Value Engine: what a home is worth

Location explains a quarter of valuation error, and map data captures most of it without knowing the district. Size, rooms and legal status alone explain 0.714 of the variation in asking prices. Adding the district cuts the valuation error by 25%. Measured distances to the centre, metro, schools and markets, without the district, cut it by 15%, about 60% of the district gain (Exhibit 12). For the business this means the engine can value homes in districts with few listings, where comparables are hardest to find.

Exhibit 12. Map data alone recovers 60% of the accuracy gain from knowing the district

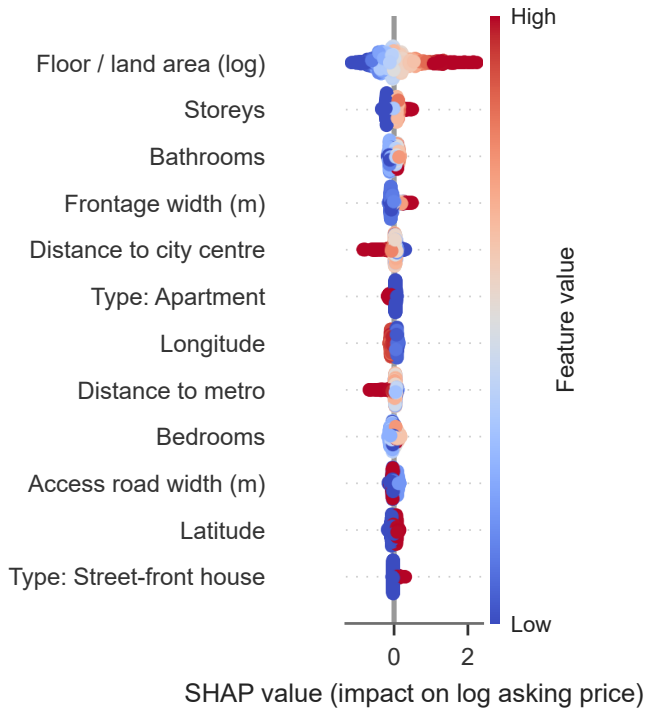
Information used	R^2	Typical error	Error cut
Size, rooms, legal status	0.714	0.560	base
+ district	0.844	0.422	25%
+ map distances, no district	0.794	0.477	15%
+ district and map distances	0.848	0.417	26%
+ type-specific size effects	0.850	0.414	26%

Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); OpenStreetMap contributors (ODbL); regression on log asking price; typical error = cross-validated RMSE on the log scale; full table in Appendix A4.

Distance to the centre costs more in Hanoi than in HCMC. Before any adjustment, a home twice as far from the centre asks about 37% less per m^2 in Hanoi and 29% less in HCMC. Property type matters as much as place. Street-front houses ask 4.5 to 4.9 times the apartment price per m^2, because their price is quoted per m^2 of land and includes business value. Accuracy must therefore be judged type by type, never as one average.

Size, height, road access and title drive the price. A 10% larger home asks about 9.5% more. Each extra storey is associated with 6.8% higher asking prices and each metre of access road width with 1.3%, holding district and amenities constant. Listings with an unknown or informal legal status ask about 10% less than those

Exhibit 13. Size and building attributes set the level; distance to the centre and to metro are the strongest location drivers



Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); OpenStreetMap contributors (ODbL); contribution of each input to the test-set estimates (SHAP values), top 12 inputs excluding district.

with a red or pink book. These are associations in asking prices, not causal effects.

The gradient-boosting model is the better engine (Exhibit 11). On 1,054 homes it had not seen, it places 53.9% of estimates within ±20% of the asking price, against 44.3% for the regression. The regression meets its target (R^2 0.850 against 0.75). The typical error of the boosting model, 0.354 on the log scale, misses the 0.20 target. Half of all estimates miss by more than 18.1%, so the engine needs a range, not a single number.

Accuracy differs by type, and that sets the launch scope. 64% of alley houses and 60% of apartments fall within ±20%. Villas reach 47%, land 43% and street-front houses 42%; listing fields capture poorly what makes a frontage valuable. The 90% range covers 90.5% of test homes as intended. It spans 96% of the estimate for alley houses but 180% for land. Expensive homes tend to be under-valued relative to cheap ones, which lenders must monitor (Appendix A2).

Area dominates the estimate, then storeys and bathrooms (Exhibit 13). Distance to the centre and to the nearest metro station are the strongest location drivers. The value is flat across the inner districts and falls beyond them. With only 20 stations in Hanoi and 14 in HCMC, metro distance mostly separates the core from the periphery. A premium for homes next to a station cannot yet be isolated.

SO WHAT FOR THE BUSINESS

Launch the engine as a pricing reference for apartments and alley houses, always with its range. Villas, land and street-front houses need a desk review until frontage and plot data improve.

Stale Listing Radar: which listings will sit unsold

The radar ranks stale risk well enough to plan a weekly review list. A listing counts as stale when it has been on the site for more than 180 days since creation. At 90 days, 82% of listings would count as stale, too many to separate, so the main specification uses 180 days (65% stale). Checks at 60, 90 and 120 days are in the appendix. The radar reaches an AUC of 0.81 (95% interval 0.78 to 0.83), against 0.74 for a logistic regression (Exhibit 14).

The top of the list is where the value is (Exhibit 15). Listings in the top 10% of scores are stale 1.5 times as often as the average listing: 99% against a base rate of 65%. Because most listings are stale, the operational flag covers the top quarter of scores, a review load of one listing in four. If a missed stale listing costs five times a false alert, the best threshold flags 87% of listings instead of 70%.

The alert threshold is a business choice, not a model setting. At the threshold that balances misses and false alerts on training data, the radar finds 85% of stale test listings, and 79% of its alerts are correct. A sales manager with a small team should use the top-quarter list. A manager who loses more from a missed stale listing than from a wasted review should lower the threshold.

Listings that sit unsold are not mainly overpriced listings. A rule that flags listings asking more than 15% above fair value scores an AUC of 0.45, no better than a coin flip. The price gap ranks 12 of 26 inputs (Exhibit 16). The top drivers are how fast other listings in the district sell, whether a registered broker posted the listing, and the property type. Older listings even ask slightly less than their fair value. A plausible reason is that they were priced in an earlier, cheaper market and never repriced.

SO WHAT FOR THE BUSINESS

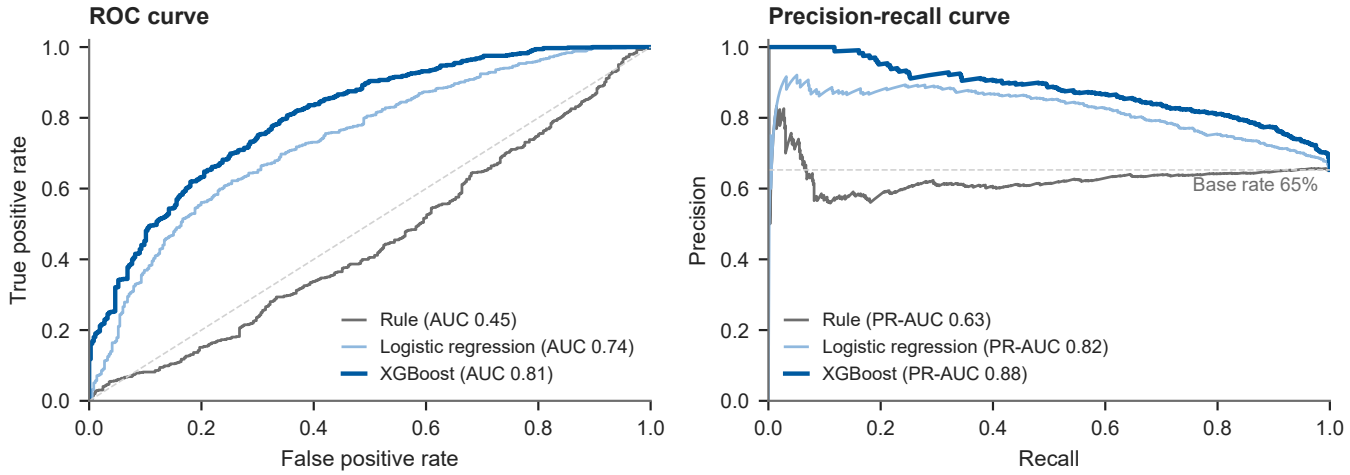
Use the radar to decide where brokers spend effort and the price gap to decide what to tell a seller. A true time-to-sale label from weekly data collection will sharpen both.

Market Map: how districts differ

Four district segments give managers four different messages. The map groups the 34 districts with at least 30 listings (97% of listings) by price level, listing age, accessibility, housing mix, price gap and volume. The four segments overlap; treat them as targeting groups, not hard borders.

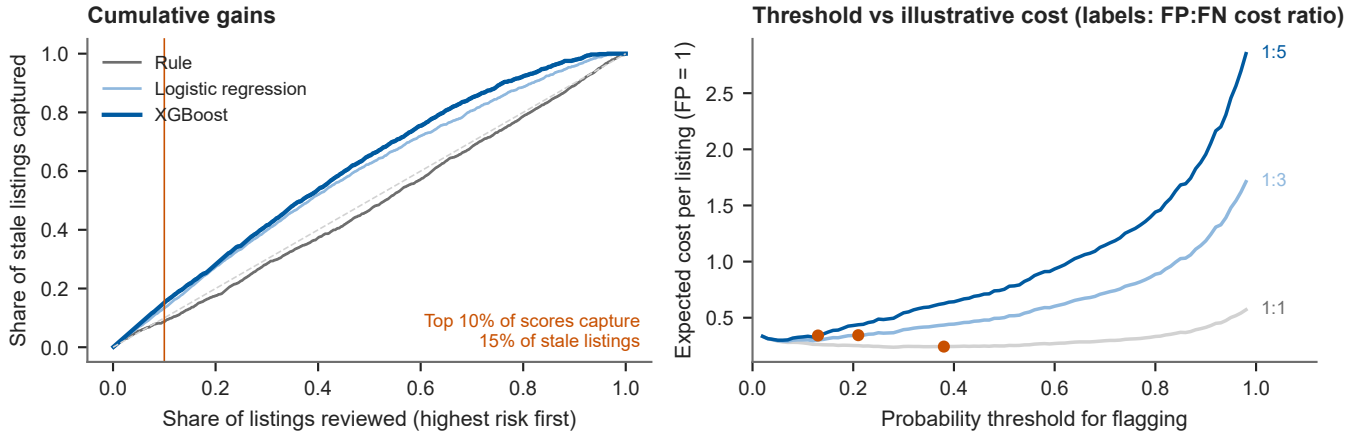
- **Prime core** (Quận 1, Quận 3, Quận Hoàn Kiếm) asks 415 million VND per m^2, 3.7 times the mid-market level, at a median 1.7 km from the centre.

Exhibit 14. The Stale Listing Radar separates stale from fresh listings with an AUC of 0.81; a rule based on the price gap does no better than a coin flip



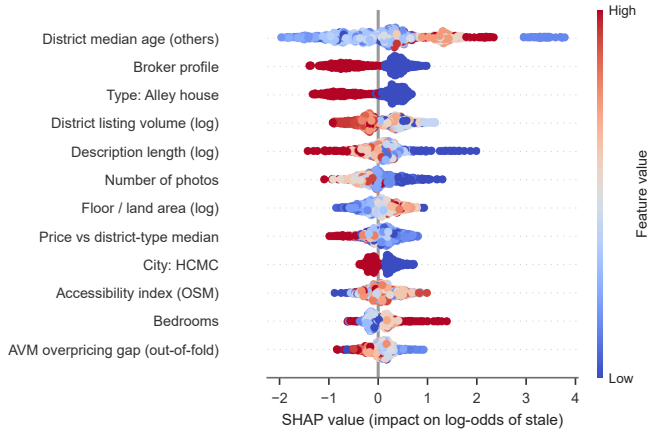
Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); 1,054 test listings; team analysis.

Exhibit 15. Listings in the top 10% of risk scores are stale 1.5 times as often as the average listing; costlier misses push the alert threshold down



Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); test listings; costs are illustrative team assumptions (a false alert costs 1 unit).

Exhibit 16. District speed and the seller's profile drive stale risk; the price gap ranks 12 of 26



Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); contribution of each input to the test-set risk scores (SHAP values).

- **Apartment mid-market** (13 districts, 2,487 listings) is the slowest segment: median listing age 364 days.
- **Townhouse mid-market** (14 districts) is house-heavy and the fastest (254 days), with the lowest stale-risk share (14%).
- **Peripheral growth** (4 Hanoi districts, 13.6 km out) has the lowest access, the highest price gap and the highest stale-risk share (36%).

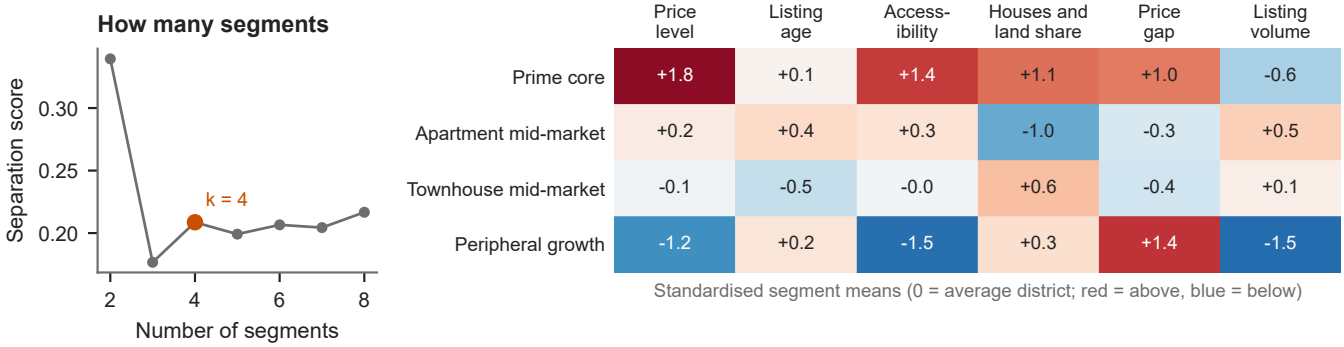
Exhibit 17 profiles the segments and Exhibit 18 shows where they sit. The two mid-market belts hold 2,487 and 2,201 listings, most of the market, so they decide broker workload. The small prime core carries the highest prices and the small periphery the highest stale risk.

> **SO WHAT FOR THE BUSINESS**
>
> Defend price in the prime core, push slow apartment-belt listings harder, use the fast townhouse belt for volume and treat the periphery as a land market. Review segment borders each quarter.

IMPACT BY DEPARTMENT

The same three outputs serve four departments, each with its own decision and KPI. The fair value with its range, the stale-risk rank and the district segment reach sales, marketing, risk and management in different forms.

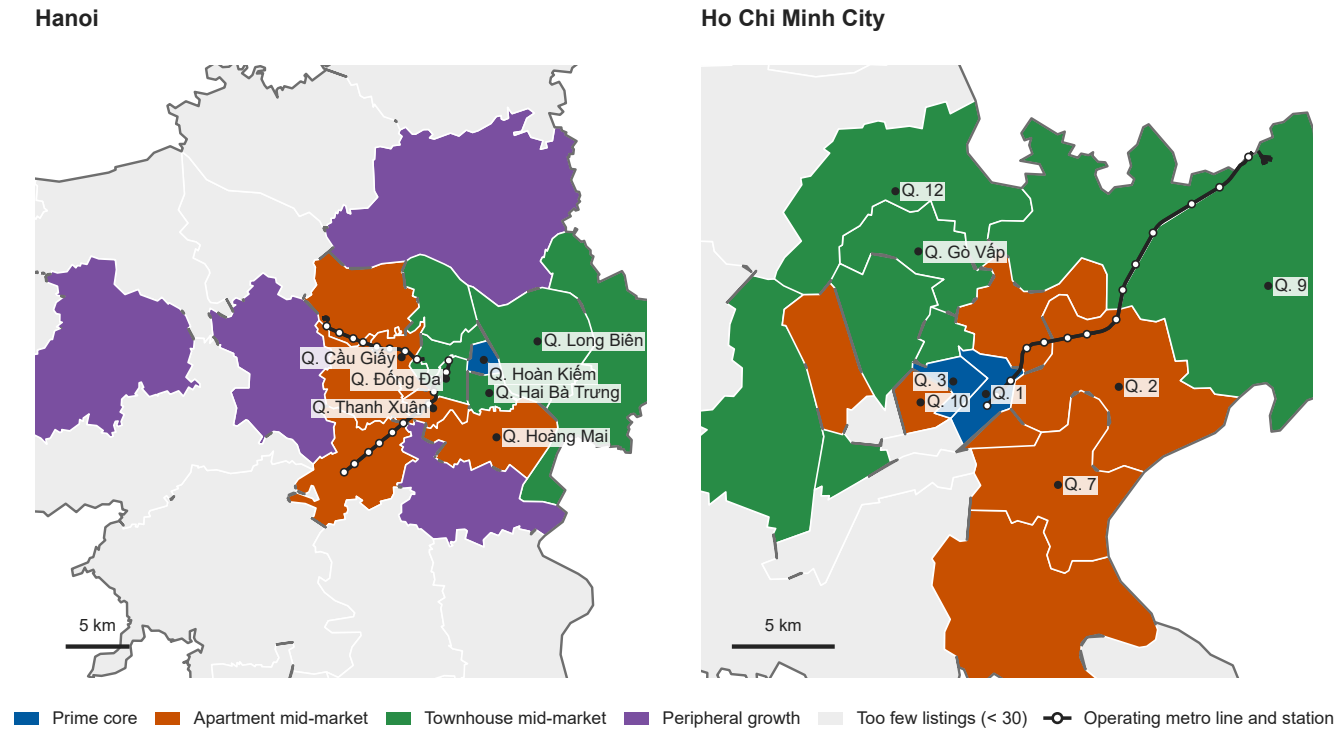
Exhibit 17. Four district segments: a small prime core, two mid-market belts that differ by housing type and speed, and a land-led periphery



Segment	Districts (HN)	Listings	Price/m² (VND m)	Age (days)	Access index	Houses, land share	To centre (km)
Prime core	3 (1)	249	415	328	+1.41	84%	1.7
Apartment mid-market	13 (6)	2,487	111	364	+0.18	46%	6.1
Townhouse mid-market	14 (6)	2,201	114	254	-0.17	78%	7.3
Peripheral growth	4 (4)	149	118	278	-1.59	71%	13.6

Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); OpenStreetMap contributors (ODbL); 34 districts with at least 30 listings; table shows medians across the districts of each segment; team analysis.

Exhibit 18. The prime core sits at the centre of each city; apartment and townhouse belts surround it, and the land-led periphery lies in Hanoi's outer districts



Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); OpenStreetMap contributors (ODbL); district boundaries: GADM 4.1 (pre-2025 districts, as listed on the site); Nhà Bè (Apartment mid-market) lies south of the HCMC frame; team analysis.

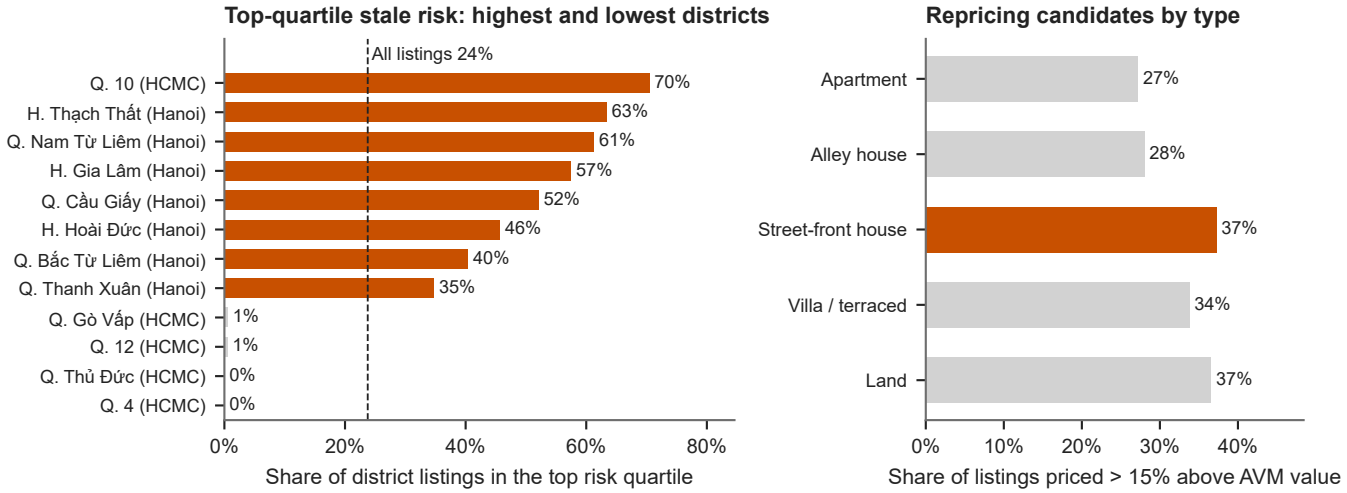
Sales and brokerage

Decision: which sellers to coach on price and which listings to prioritise. **Evidence:** the share of listings in the top quarter of stale risk ranges from 70% in Quận 10 (HCMC) to 0% in Quận 4 (HCMC) (Exhibit 19). Repricing candidates, listings asking more than 15% above fair value, are most common for street-front houses (37%) and land (37%), against 27% for apartments. **Talking point:** in high-risk districts, open the price conversation early with the range and comparables. **KPI:** median days to sale, coached against uncoached listings.

Marketing and research

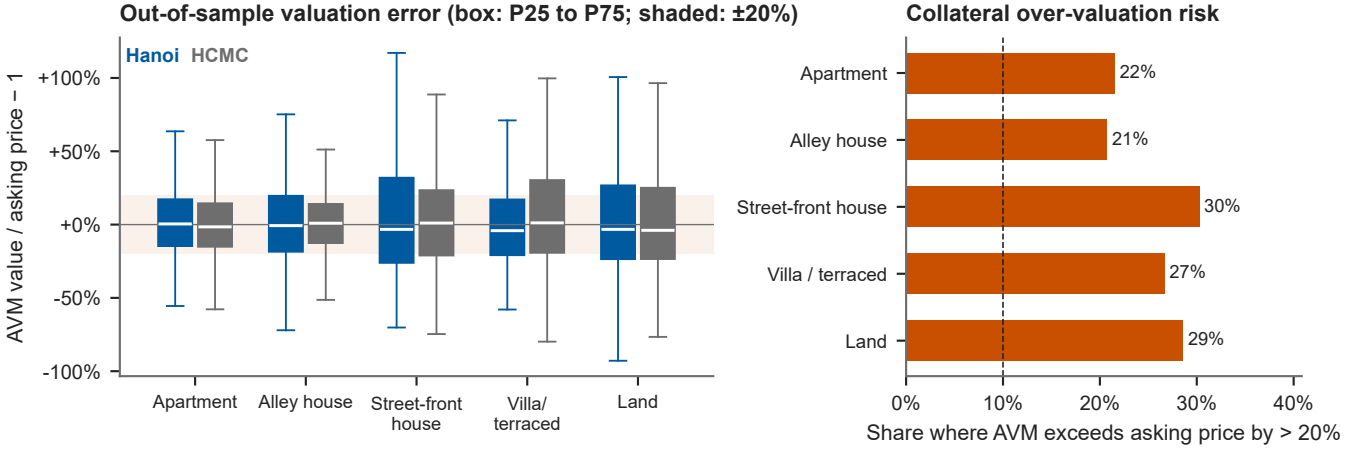
Decision: which areas to target and with what message. **Evidence:** the four segments differ in price level, speed and housing mix (Exhibit 17). **Messages:** scarcity and status in the Prime core; value and amenities in the slow Apartment mid-market, where listings need the most support. Speed and choice suit the Townhouse mid-market; land and future infrastructure suit the Periph-

Exhibit 19. Sales: stale risk concentrates in a few districts, and more than a third of street-front and land listings ask over 15% above fair value



Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); fair values and risk scores estimated on listings the models had not seen; districts with at least 30 listings; team analysis.

Exhibit 20. Risk: no property type is yet safe for collateral checks without a desk review; street-front houses and land carry the most over-valuation risk



Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); fair values estimated on listings the models had not seen; team analysis.

eral growth. **KPI:** leads per listing by segment and campaign.

Risk and credit

Decision: when a fair value can pre-screen collateral before a manual appraisal. **Evidence:** the engine values 25% of listings more than 20% above their asking price (Exhibit 20). At a 70% loan-to-value ratio such a case would lend 84% of the asking price. Over-valuation is least frequent for alley houses (21%) and apartments (22%) and most frequent for street-front houses (30%) and land (29%). **Rule:** fair value plus desk review for apartments, alley houses and villas; manual appraisal for street-front houses and land. **KPI:** share of pre-screens confirmed within ±10% by the final appraisal.

Executive management

Decision: where to open offices, recruit brokers or commit capital. **Evidence:** Exhibit 21 places every district by speed and price level. Districts to the left sell fastest; those in the upper left combine high prices with speed and deserve senior broker capacity. **KPI:** quarterly change in median listing age by segment.

Every engine has a business owner (Exhibit 23). The data team runs all three engines. Each department owns the decision its engine feeds, which ties every model change to a business owner.

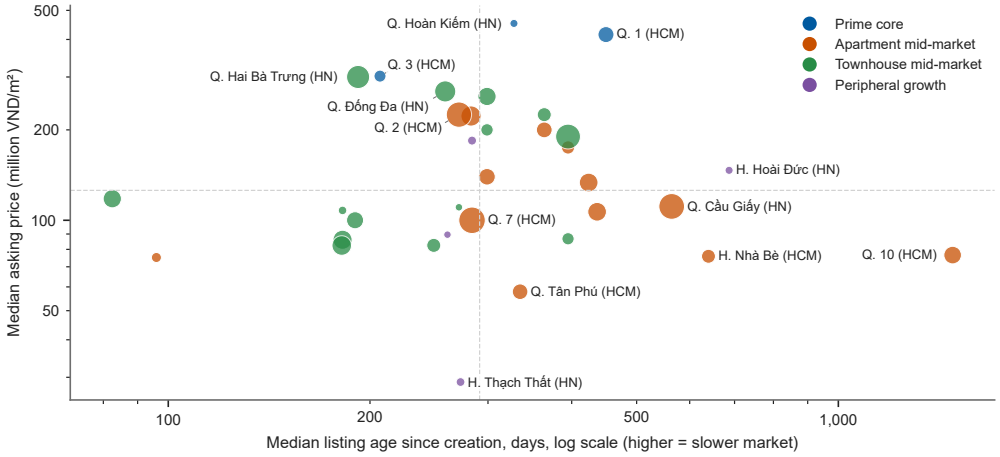
TOOLS AND WORKFLOW

Open-source tools keep the prototype cheap and the path to production short. Each tool was chosen on three tests: does it scale to the full market, can a team work on it together, and can a manager use the output without code.

The prototype runs on a laptop. One automated pipeline collects the listings, adds the map data, re-trains the models and refreshes every exhibit. In production the same pipeline moves to a cloud database with spatial queries, runs weekly, and keeps every model version with its accuracy record. A standalone browser version of the dashboard accompanies this report.

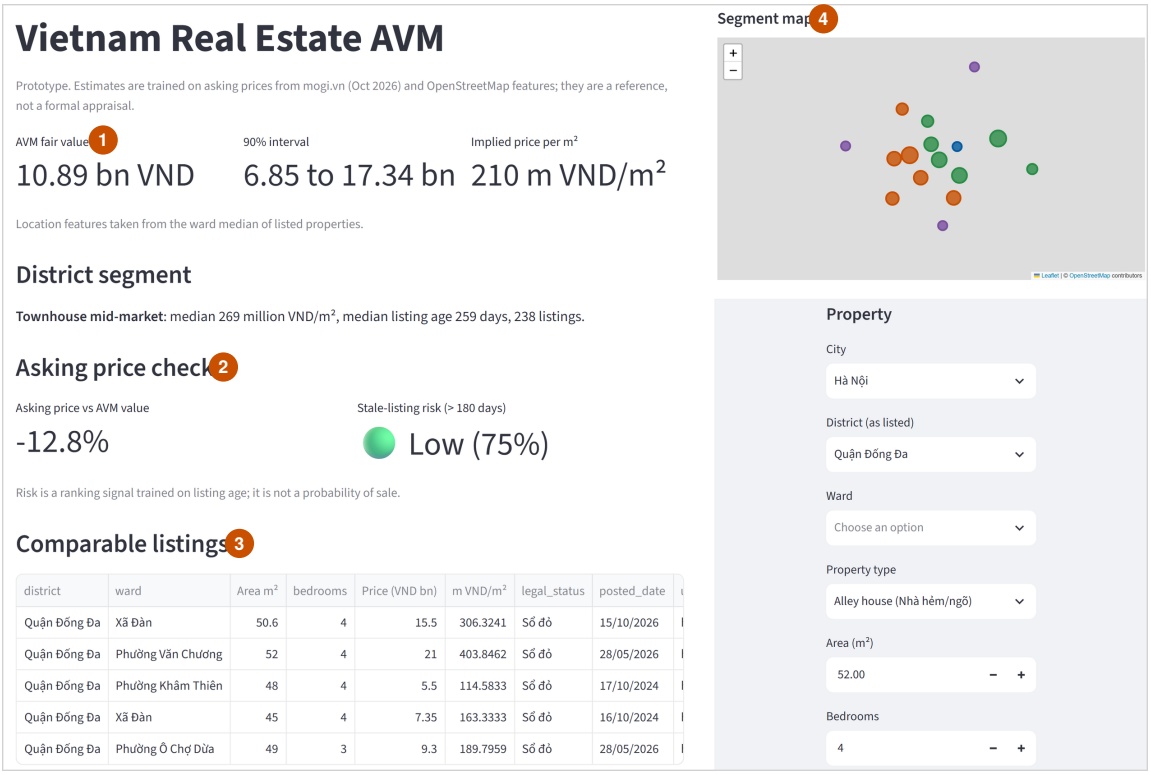
From data refresh to a manager's decision in four steps.

Exhibit 21. Management: fast, high-priced districts sit in the upper left and deserve the most broker capacity



Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); OpenStreetMap contributors (ODbL); bubble size = listings; dashed lines = medians across districts; team analysis.

Exhibit 22. A broker enters a home and an asking price and reads the fair value, its range, the price gap, the stale-risk light, comparables and the district segment



Source: dashboard prototype with the engines trained on mogi.vn sale listings, 15 October 2026 (N = 5,268). Callouts: (1) fair value and 90% range; (2) asking-price gap and stale-risk light; (3) five comparable listings; (4) district segment map. Inputs (right, below the map) are entered in the sidebar.

Exhibit 23. Each department owns one decision and uses at least two engines

Department	Fair value	Stale radar	Market Map
Sales and brokerage	R	A, R	I
Marketing and research	C	I	A, R
Risk and credit	A, R	C	I
Executive management	I	I	A
Data team	R	R	R

Source: team design. R = responsible (uses daily), A = accountable (owns the decision), C = consulted, I = informed.

(1) The weekly collection and cleaning job updates the property table. (2) The engines re-score every listing, and re-train monthly if accuracy on new listings falls below the launch level. (3) The dashboard and the district brief refresh automatically. (4) A broker enters a home and an asking price and reads the fair value, its range, the gap, the stale-risk light and five comparables (Exhibit 22).

Governance keeps the numbers trustworthy.

- **Change control.** A new model version goes live only after its business owner (Exhibit 23) signs off its accuracy report by type.
- **Monitoring.** A monthly report tracks accuracy by type, stale flags and broker usage.
- **Access.** Brokers see estimates; only the data team sees raw listings; no personal data is stored.

Exhibit 24. Each layer uses a tool that scales, supports teamwork and needs no code from managers

Layer	Tool	Why it fits
Collection	Python	Resumes after interruptions; 7,792 listings an hour with polite pacing
Map data	OpenStreetMap	Free, open licence; one download per city each quarter
Storage	Files now; cloud database later	No setup for the prototype; spatial queries and access control in production
Models	Regression, XGBoost, SHAP	A transparent baseline plus an accurate model, both explainable
Interface	Streamlit dashboard	A valuation in a browser, no code needed
Reporting	LaTeX	Figures and numbers generated automatically from the analysis
Teamwork	Git	Shared history; one automated rebuild of data, models and reports

Source: team design.

Exhibit 25. Base case: partner value of VND 11.9 bn a year leaves net value of 9.7 bn, 4.4 times the run cost; the conservative case does not cover its cost

Input	Cons.	Base	Opt.
Brokers using the tool	100	300	500
Valuations per broker a month	10.0	20.0	25.0
Hours saved per valuation	0.50	1.00	1.25
Broker cost (VND m/hour)	0.10	0.15	0.20
Mortgage files pre-screened a year	1,000	3,000	6,000
Estimates within ±10% (PPE10)*	30%	30%	30%
PPE10 files where the bank waives appraisal	30%	50%	70%
Cost of an appraisal (VND m)	1.50	2.50	4.00
Run cost, 3-person team (VND m/year)	3,000	2,200	1,600
Buyer deals guided a year	300	1,000	2,500
Listings above the 90% range*	4%	4%	4%
Median excess above the range*	12%	12%	12%
Share of excess negotiated away	10%	20%	30%
Median asking price (VND bn)*	9.60	9.60	9.60
Annual value, VND bn	**Cons.**	**Base**	**Opt.**
Broker time saved	0.6	10.8	37.5
Bank appraisal cost avoided	0.1	1.1	5.0
Run cost	3.0	2.2	1.6
Net partner value	**−2.3**	**9.7**	**40.9**
ROI (net value / run cost)	**−0.8x**	**4.4x**	**25.6x**
Buyer overpayment avoided (not in ROI)	1.6	10.4	39.0

Source: * measured in this analysis; all other inputs are team assumptions; ROI = (partner value − run cost) / run cost.

Exhibit 24 lists the tool behind each layer and why it fits.

VALUE AND ROI

The base case pays: net partner value of VND 9.7 billion a year, 4.4 times the run cost. Partners gain VND 11.9 billion a year against a run cost of VND 2.2 billion (Exhibit 25). The figures are scenario estimates on labelled assumptions, not measured outcomes. Three levers create the value:

- **Broker time saved:** an instant estimate and range replace a manual search for comparables.
- **Bank appraisal cost avoided:** a partner bank waives part of the full appraisals where the estimate falls within ±10% (29.8% of homes).
- **Buyer overpayment avoided:** only listings above the 90% range count (4% of listings, median excess 12%), so model error is not counted. This value goes to buyers, outside the ROI.

Adoption, not accuracy, decides the case (Exhibit 26). With 100 brokers making 10 valuations a month and saving half an hour each, the platform loses VND 2.3 billion a year. The number of brokers using the tool moves net value more than any other input.

The strategic value goes beyond money. It would be the first public valuation of all property types in the two cities. The weekly listing panel becomes a price index and time-on-market record that competitors cannot copy quickly.

ROADMAP, RISKS AND RECOMMENDATIONS

Twelve months, three phases, each closed by a measured gate (Exhibit 27). Exhibit 28 lists the five risks and their mitigation.

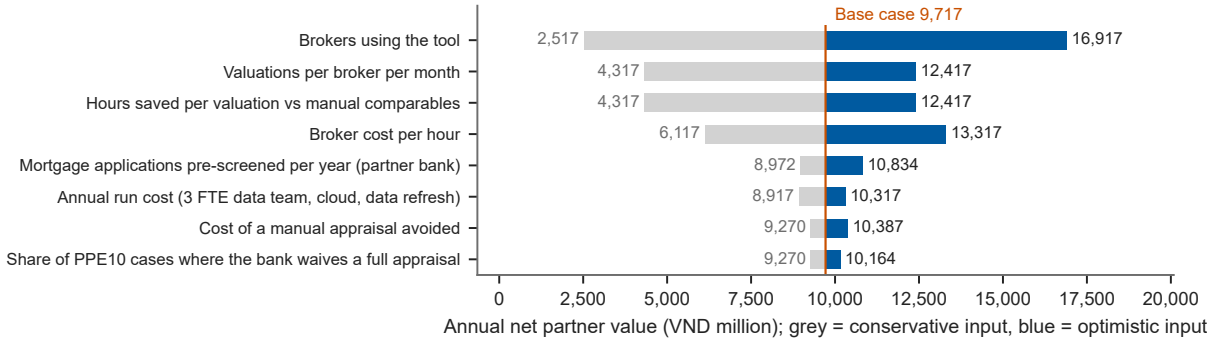
Each gate is a number the steering group can check.

- **Gate 1, month 3 (continue to scale?).** On pilot deals, apartments and alley houses stay at or above the test hit rate of 60% within ±20%. Pilot brokers use the tool weekly. If accuracy drops, the pilot runs one more quarter.
- **Gate 2, month 7 (start lending use?).** The partner bank compares fair values with its own appraisals on live files and agrees a waiver rule for files within ±10%. Active brokers reach the 300 target.
- **Gate 3, month 12 (launch publicly?).** Accuracy by type is published and stable for three monthly reports in a row. Legal review of the disclaimers is complete.

Recommendations

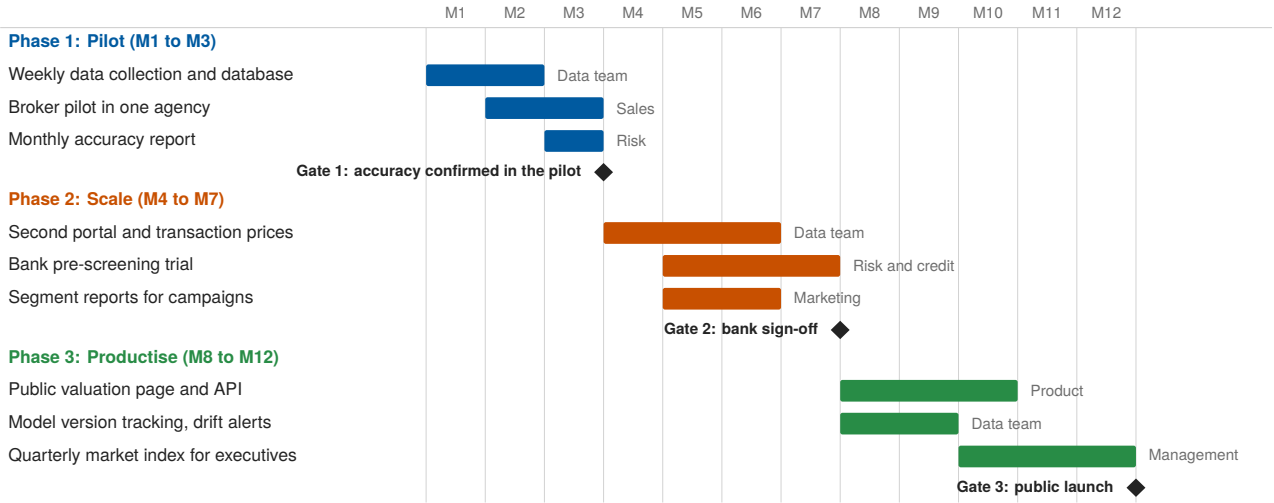
1. **Launch the Fair Value Engine to brokers for apartments and alley houses first.** These types reach 60% and 64% within ±20% on unseen homes. Show every estimate with its 90% range and five comparables.
2. **Keep street-front houses, villas and land on desk review.** Their hit rates (42%, 47%, 43%) are too low for automated use. Add frontage quality, plot shape and business-use fields to the listing form.
3. **Sell the Stale Listing Radar as a priority list, not an overpricing detector.** Its top 10% of listings are stale 1.5 times as often as average, which justifies a weekly review list. Base repricing advice on the price gap from the Fair Value Engine.
4. **Build a true time-to-sale label in Phase 1.** Weekly data collection shows when listings disappear. Within three months this replaces the creation-age proxy.
5. **Calibrate on transactions before any lending use.** Before the bank trial, re-measure accuracy against appraisal or notarised prices.
6. **Make broker adoption the first business KPI.** The case turns positive only with broad adoption. Target 300 active brokers by the end of Phase 2 and report usage next to accuracy every month.

Exhibit 26. Broker adoption drives the business case more than any accuracy metric



Source: team assumptions in Exhibit 25; each bar varies one input between its conservative and optimistic value, holding the others at base.

Exhibit 27. Twelve months in three phases, with a go or no-go gate at the end of each phase



Source: team plan; owners refer to the departments in Section 6.

Exhibit 28. Five risks, each with a mitigation; the asking-price gap and the stale label matter most for the current results

Risk	L / I	Mitigation
Listing site changes its pages or blocks collection	M / H	Tested page readers; adaptive pacing; second portal in Phase 2; negotiate a data feed
Asking prices differ from transaction prices	H / H	Calibrate on bank appraisals or notarised prices in Phase 2; present results as asking-price references until then
Stale label is noisy (refreshed dates, unseen sales)	H / M	Creation-date labels now; delisting events from weekly collection give a true time-to-sale label
Accuracy drifts as prices move	M / M	Monthly accuracy report by type; re-train when PPE20 falls 5 points below launch
Legal and reputational risk of publishing valuations	L / H	Ranges, not single numbers; disclaimers; no personal data; published method and accuracy

Source: team analysis. L / I = likelihood / impact: H high, M medium, L low.

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TECHNICAL APPENDIX

Design safeguards

1. **No listing age among the radar's inputs.** The stale label is defined by listing age, so age as an input would reveal the answer (target leakage). Listing age, posted, updated and photo dates, the listing ID and anything derived from them are excluded.
2. **Price gap estimated on unseen listings.** The gap between asking price and fair value used by the radar comes from five-fold cross-validated predictions for training rows (out-of-fold) and from the held-out model for test rows. An in-sample gap would carry the training fit into the classifier.
3. **Listing age from creation, not from the posted date.** The site assigns listing IDs in creation order. IDs are only weakly related to posted dates (Spearman $\rho = 0.44$) but track first photo-upload dates almost exactly ($\rho = 0.993$). Also, 59% of listings show a posted date more than 30 days after their photos were uploaded. The label therefore uses the creation date implied by the listing ID, calibrated to photo dates with an isotonic fit. The median listing age is 90 days from the posted date, 297 days from the first photo and 299 days from the listing ID.
4. **No poster account age among the radar's inputs.** An account cannot post before it exists, so account age bounds listing age mechanically; it is excluded for the same reason as item 1.

The accessibility index is the first principal component of six inverted distances (metro, school, hospital, park, market, centre); it explains 54% of their variance, with loadings between 0.35 and 0.46.

Accuracy by city and type

The gradient-boosting model has a median absolute error of 18.1%, PPE10 of 29.8% and a coefficient of dispersion of 27.0. Its price-related differential of 1.12 lies above the 0.98 to 1.03 range used in ratio studies (International Association of Assessing Officers 2013). Split-conformal intervals calibrated on out-of-fold errors by type cover 90.5% of test homes (Exhibit 29).

Exhibit 29. Gradient boosting beats the regression in most city and type cells; land and street-front houses carry the widest errors

City	Type	N	RMSE (log)		PPE20		MdAPE		COD
			OLS	XGB	OLS	XGB	OLS	XGB	XGB
Hanoi	Villa / terraced	55	0.343	0.292	47%	49%	21.3%	21.1%	23.0
Hanoi	Apartment	126	0.312	0.243	50%	59%	19.9%	16.8%	20.2
Hanoi	Alley house	156	0.340	0.307	46%	55%	21.4%	18.6%	22.8
Hanoi	Street-front house	100	0.527	0.431	22%	38%	38.7%	27.6%	36.2
Hanoi	Land	90	0.537	0.564	38%	37%	26.9%	27.7%	47.0
HCMC	Villa / terraced	63	0.423	0.339	48%	46%	22.6%	23.0%	26.7
HCMC	Apartment	149	0.399	0.344	49%	62%	20.4%	15.1%	26.9
HCMC	Alley house	144	0.309	0.234	60%	73%	15.6%	12.6%	16.5
HCMC	Street-front house	86	0.443	0.330	31%	48%	34.0%	21.8%	26.7
HCMC	Land	85	0.593	0.435	41%	51%	24.6%	16.4%	33.7

Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); 1,054 held-out test homes.

Stale Listing Radar and Market Map checks

Exhibit 30 compares the radar with a price-gap rule and a logistic regression on the test set. With the posted-date label the AUC is near 1, because that label mainly records whether a broker account keeps re-posting the listing (Exhibit 31). This is why the creation-date label is preferred. **Market Map.** Silhouette scores are low at every k; within the preferred range of 3 to 5 segments, $k = 4$ scores best (0.21, against a target of 0.45). Ward hierarchical clustering with the same k agrees with K-Means at an adjusted Rand index of 0.47. Features are standardised and winsorised at ± 2.5 standard deviations; listing age enters as a log median.

Exhibit 30. XGBoost ranks stale listings best; the price-gap rule is no better than chance

Model	AUC	95% CI	PR-AUC	Brier	Top-decile precision	Top-decile capture
Rule: AVM gap > +15%	0.45	0.41 to 0.49	0.63	–	57%	9%
Logistic regression	0.74	0.71 to 0.77	0.82	0.189	87%	13%
XGBoost	0.81	0.78 to 0.83	0.88	0.176	99%	15%
Base rate (share stale in the test set): 65%; N test = 1,054						

Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); test set; label: on site more than 180 days since creation; Brier = mean squared error of the predicted probability.

Exhibit 31. Ranking quality holds across creation-age thresholds; the posted-date label is predictable for the wrong reason

Age measure	Days	Model	Positive share	AUC	95% CI
Creation age (listing ID)	180	Rule (AVM gap > 15%)	65%	0.45	0.41 to 0.49
		Logistic regression	65%	0.74	0.71 to 0.77
		XGBoost	65%	0.81	0.78 to 0.83
Creation age (listing ID)	60	Rule (AVM gap > 15%)	89%	0.51	0.46 to 0.57
		Logistic regression	89%	0.83	0.80 to 0.86
		XGBoost	89%	0.88	0.84 to 0.91
Creation age (listing ID)	90	Rule (AVM gap > 15%)	82%	0.46	0.42 to 0.50
		Logistic regression	82%	0.82	0.79 to 0.85
		XGBoost	82%	0.85	0.82 to 0.88
Creation age (listing ID)	120	Rule (AVM gap > 15%)	77%	0.46	0.42 to 0.50
		Logistic regression	77%	0.81	0.78 to 0.85
		XGBoost	77%	0.85	0.82 to 0.88
Posted date (refreshed)	60	Rule (AVM gap > 15%)	55%	0.47	0.44 to 0.51
		Logistic regression	55%	0.96	0.95 to 0.98
		XGBoost	55%	0.99	0.98 to 0.99
Posted date (refreshed)	90	Rule (AVM gap > 15%)	50%	0.48	0.45 to 0.52
		Logistic regression	50%	0.91	0.90 to 0.93
		XGBoost	50%	0.96	0.95 to 0.97

Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); held-out test sets; 95% intervals from 1,000 (main) or 200 (robustness) bootstrap resamples.

Full regression table, model settings and data dictionary

Exhibit 32 gives the full regression ladder behind Exhibit 12. Exhibit 33 lists the gradient-boosting settings, and Exhibit 34 the fields the engines use.

Exhibit 32. Hedonic regression ladder on log asking price

	(1) Physical	(2) + District FE	(3) + OSM spatial	(4) + Interactions
Log area	0.931***	0.935***	0.929***	0.954***
	(0.046)	(0.025)	(0.025)	(0.057)
Bedrooms	0.016	−0.007	−0.011	−0.017**
	(0.012)	(0.008)	(0.008)	(0.008)
Bathrooms	−0.002	0.018*	0.022**	0.025***
	(0.013)	(0.010)	(0.010)	(0.009)
Storeys	0.077***	0.064***	0.065***	0.066***
	(0.013)	(0.010)	(0.010)	(0.008)
Frontage (m)	0.010	0.005	0.004	0.006*
	(0.007)	(0.004)	(0.004)	(0.004)
Access road width (m)	0.016***	0.013***	0.012***	0.013***
	(0.005)	(0.004)	(0.004)	(0.004)
Legal: sale contract	0.146	0.143*	0.144*	0.151**
	(0.091)	(0.086)	(0.078)	(0.076)
Legal: other/unknown	0.067	−0.090**	−0.097***	−0.109***
	(0.064)	(0.039)	(0.035)	(0.042)
Log distance to CBD			−0.053	−0.047
			(0.052)	(0.053)
Log distance to metro			−0.026	−0.032
			(0.035)	(0.032)
Log distance to school			−0.009	−0.006
			(0.024)	(0.024)
Log distance to hospital			−0.034	−0.028
			(0.028)	(0.024)
Log distance to park			−0.054**	−0.055**
			(0.022)	(0.023)
Log distance to market/mall			−0.059**	−0.055**
			(0.025)	(0.026)
Log distance to lake/river			−0.015	−0.012
			(0.023)	(0.023)
Bus stops within 500 m			−0.004	−0.004
			(0.004)	(0.003)
District fixed effects	No	Yes	Yes	Yes
Type × city, type × log area	No	No	No	Yes
N	5,268	5,268	5,268	5,268
R^2	0.714	0.844	0.848	0.850
Adjusted R^2	0.713	0.842	0.846	0.848
5-fold CV RMSE (log)	0.560	0.422	0.417	0.414

Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); OpenStreetMap contributors (ODbL). Standard errors clustered by district in parentheses; *, **, ***: $p < 0.10$, 0.05, 0.01. All columns include type and city dummies and missing-value flags; reference legal status: red or pink book (sổ đỏ/sổ hồng). Without district effects, physical and map features give R^2 = 0.794 and CV RMSE = 0.477. CBD distance is not statistically different from zero within districts ($p = 0.305$); sale-contract listings ask more ($p = 0.048$).

Exhibit 33. Gradient-boosting settings, tuned by cross-validation on the training set only

Setting	Fair Value Engine	Stale Listing Radar
Tree depth	5	6
Number of trees	2,284	1,115
Learning rate	0.02	0.02
Share of rows per tree	0.79	0.88
Share of features per tree	0.54	0.67
Minimum leaf size	4.00	1.00
Shrinkage penalty	1.24	0.57
Weight on stale listings	none	0.53
Search	40 settings, 5-fold	30 settings, 5-fold

Source: team analysis.

Exhibit 34. Data dictionary: the fields the engines use

Field	Description	Source	% missing
Asking price	Total asking price in VND (what the Fair Value Engine predicts)	Listing	0
Property type	Apartment, alley house, street-front house, villa or terraced, land	Listing	0
City, district, ward	Location as listed on the site (pre-2025 district names)	Listing	0
Map point	Latitude and longitude from the listing map	Listing	0
Area	Floor area (apartments) or land area (houses, villas, land), m^2	Listing	0
Bedrooms, bathrooms	Room counts	Listing	0
Storeys	Number of floors, often read from the description	Listing text	54
Frontage	Plot width facing the street or alley, m	Listing	29
Plot depth	Plot depth, m	Listing	31
Access road width	Width of the alley or street in front, m	Listing text	87
Legal status	Red or pink book, sale contract, other or unknown	Listing	0
Text flags	Car access, corner lot, business use, lake view, near school, furnished, owner selling, urgent sale	Listing text	0
Seller profile	Registered broker profile or private account	Listing	0
Photos, description length	Number of photos; length of the description	Listing	0
Listing age	Days on the site since creation (label of the Stale Listing Radar only)	Derived	0
Distances	Distance to the centre, nearest metro station, school, hospital, park, market, water	Map data	0
Counts nearby	Bus stops, schools and other points within 500 m and 1 km	Map data	0
Accessibility index	One score combining six distances (higher = closer to amenities)	Map data	0

Source: mogi.vn sale listings, 15 October 2026 (N = 5,268); OpenStreetMap contributors (ODbL). Percent missing is measured before imputation.